CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) State TWO (2) facts about how a chain of trust is established in the Trusted Computing architecture, referring to the role of the TPM and the sequence of measured components.

(2 marks)

b) Discuss the NGSCB/Nexus architecture by describing its two primary system components and the security functions each performs within the trusted platform.

(6 marks)

QUESTION 2

a) Explain the two stages of the biometric authentication process and state what activities occur in each stage.

(2 marks)

b) State six facts about biometric technologies used for authentication, covering the definition of a biometric, the two categories of biometric applications, and the specific technologies and their characteristics.

(6 marks)

QUESTION 3

a) State TWO (2) differences between morals and laws as forms of conduct guidance for computing professionals.

(2 marks)

b) Explain the role of a professional code of ethics and conduct for computing professionals, referring to the ACM code and its professional conduct expectations.

(6 marks)

QUESTION 4

a) State TWO (2) characteristics of a Trojan horse as a type of malware.

(2 marks)

b) State SIX (6) techniques by which malware propagates itself or conceals its activity within an infected system, as described in the software security lecture.

(6 marks)

QUESTION 5

a) Explain how a buffer overflow arises in a program such as the one using strcpy() into a fixed-size local buffer, and how an attacker can exploit it.

(2 marks)

a) Explain six security mechanisms and structures that Windows provides for managing user accounts, subjects and system resources, and note why the complexity of these mechanisms can itself become a security concern.

(6 marks)

QUESTION 6

a) [THEORY PLACEHOLDER - attacks/malware, 2 marks, List. No sourced question passed verification.]

(2 marks)

b) The Samy worm spread through MySpace by exploiting cross-site scripting. Describe the Samy worm case study by explaining how the worm injected itself, how it propagated to other users, and the scale of the resulting infection.

(6 marks)

QUESTION 7

a) List TWO (2) typical locations where IDS sensors are placed in a network-based IDS architecture.

(2 marks)

b) Refer to your answer in (a). Explain how preventive network security mechanisms, such as firewalls, proxy servers, network segmentation and remote access VPN, reduce the attacks that an IDS must later detect, and state the limitation of these preventive mechanisms.

(6 marks)

QUESTION 8

a) Explain two ways in which a professional code of ethics and conduct, such as the ACM code, guides the ethical conduct of computing professionals and shapes the ethical impact of the technology they build.

(2 marks)

b) Compare morals and laws as forms of conduct guidance for computing professionals, and explain how organizational responsibility for privacy is reflected in professional conduct expectations such as those of the ACM code. Provide six distinct points.

(6 marks)

QUESTION 9

a) Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

(2 marks)

b) Discuss six points about the access control matrix, access control lists and capabilities as access control structures, covering what each structure represents and the strengths or weaknesses of each.

(6 marks)

QUESTION 10

a) Explain two password security controls that can be applied to password files and user accounts to protect an authentication system.

(2 marks)

b) Describe six facts about single sign-on and Kerberos as authentication mechanisms, covering the motivations for single sign-on, the categories of solutions, the design criteria of Kerberos, and how a user's password is used at login.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Discuss how intrusion detection systems support the detection of and defence against attacks on computer systems. In your answer, describe five distinct capabilities or limitations of intrusion detection systems.

(5 marks)

b) Explain the four classes of threats to computer systems and, for each class, identify the security service that counters it, as described in the introduction to computer security lecture.

(5 marks)

QUESTION 2

A hospital is reviewing how its information systems decide whether a clinician may open a patient record. The review team wants to know what the access control subsystem is trying to achieve and which component is responsible for making each authorization decision before any record is released.

a) State five points about the goals of access control and the role of the reference monitor in authorization, as presented in the lecture.

(5 marks)

b) The hospital review team is considering formal security models for controlling access to patient records. Explain five points about the Bell-La Padula, Biba and Chinese Wall models, covering the security concern each model addresses and the access rules that each model imposes.

(5 marks)

END OF QUESTION PAPER